Dominik Szewczyk

Data Scientist · Machine Learning & AI

+48 797 788 847 · dominikszewczyk123@gmail.com · linkedin.com/in/dominik-szewczyk · dominikszewczyk224180.github.io · Silesia, Poland · open to relocation



PROFILE

Data Science and AI graduate (cum laude) who builds machine learning systems end to end, from computer vision and NLP models to dashboards and cloud deployment on Azure. Two industry internships and applied research across logistics, plant science, and urban planning, with a habit of shipping measurable results.

// 01 Experience

Data Scientist, Graduation Internship · KapiteinLabs — 2026

- Sole data scientist on a six-month graduation project, taking a predict-and-optimise system from idea to a deployed **no-code Streamlit dashboard**.
- Modeled a plasma reactor's ozone output from its physical settings (geometry, voltage, repetition rate, flow, pulse width) using **438 measurements** from TU Eindhoven.
- Added an inverse optimiser that recommends the settings needed to hit a target output, replacing slow one-off lab experiments with instant estimates.

Data Scientist, Internship · De Rooy Transport & Logistics — Feb - Jun 2025

- Built a **multi-model pipeline** forecasting where, when, and how large unexpected pickups will be (over 82% of daily assignments), wired into the planners' PTV route software.
- Pickup occurrence: a Keras model over 5 km grid cells reaching **90.2% accuracy** and 86.5% F1; location via random forest with an **18 m mean error** and 100% of predictions within 1 km.
- Engineered spatial, temporal, and holiday features for peak robustness. Impact: up to **€1,500, 1,500 km, and 945 kg CO_2 saved per day**. Featured in the LSB whitepaper.

// 02 Selected Projects

AI 3D Digital Twin & Object Removal — Antea Group & Kavel 10 · 2024 to 2025

Reconstructed urban environments from aerial and drone imagery for city planning. Benchmarked DeepFill v2, LaMa, and PowerPaint for object removal and U-Net, DeepLabV3, and FCN for segmentation, then generated 3D meshes with Instant NeRF and a VR walkthrough in Unreal Engine.

Computer Vision · NeRF · Semantic Segmentation · 3D Reconstruction

Plant Analysis with Computer Vision & Robotics — NPEC · 2023 to 2024

Segmented seeds, shoots, and roots from plant images with a U-Net and instance segmentation, localising root tips to within roughly 1 mm. Simulated a PID-controlled liquid-handling robot and applied reinforcement learning to dispense precisely at the root tips.

Computer Vision · U-Net · Reinforcement Learning

ATHENA: MLOps for Plant Science — NPEC · 2024

Took research models to production for ATHENA, a Python package and cloud app for organ segmentation and landmark detection on plant images. Built Azure pipelines with versioned assets, containerised with Docker, and served via FastAPI. Recognised publicly by NPEC.

Microsoft Azure · Docker · FastAPI · MLOps

Emotion Classification with NLP — Banijay Media · 2024

Detected emotions in video content by classifying text extracted from clip fragments. Compared Naive Bayes, logistic regression, HMM, CRF, RNN, and LSTM, then fine-tuned BERT to reach an F1 above 0.88 on the held-out test set.

NLP · Transformers (BERT) · Feature Engineering

// 03 Education

Breda University of Applied Sciences · Data Science & Artificial Intelligence — 2022 - 2026

Graduated cum laude. Two industry internships and applied research in ML, computer vision, NLP, and MLOps.

II LO im. Andrzeja Frycza Modrzewskiego · Extended Mathematics — 2019 - 2023

Top-ranked secondary school in the Silesia region.

// 04 Skills & Tools

MACHINE LEARNING & AI	Machine Learning, Deep Learning, Computer Vision, NLP & Transformers, Predictive Modeling
LANGUAGES & TOOLS	Python, scikit-learn, TensorFlow, Keras, XGBoost, FastAPI, Streamlit, Power BI, R, Git
CLOUD & MLOPS	Microsoft Azure, Azure ML, Docker, CI/CD pipelines
WAYS OF WORKING	Agile (Scrum), data visualisation, stakeholder communication
LANGUAGES	English (fluent) · Polish (native) · German (basic)
INTERESTS	Football, gym, chess (3rd category), half marathon (1:43)